

HIBA VALIYAKATH SAEREEF

AI Engineer & Edge AI Specialist

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Summary

Fourth-year CS & Data Science student experienced in building production-ready Edge AI systems that run efficiently on standard hardware. Combines deep learning research with a track record of shipping deployable, full-stack web applications. Passionate about helping Japanese SMEs automate workflows and expand globally through cost-effective, local-inference AI and computer vision tools.

Research Experience

Research Intern — IIIT Kottayam

May 2026 – Present

Satellite Disaster Assessment: MS-DCAFNet++ | PyTorch, xBD, Remote Sensing

Thrissur, India

- Designing **MS-DCAFNet++**, a novel Siamese encoder-decoder for pixel-wise building damage segmentation across five severity classes on bi-temporal satellite image pairs.
- Proposed **DCAF (Damage-Conditioned Adaptive Fusion)**: an asymmetric attention module where the post-disaster image queries pre-disaster context, solving a fundamental gap in prior symmetric models (CBAM, SE-Net).
- Replaced max-pooling with **Haar wavelet downsampling** and engineered a **hybrid Focal-Dice-Contrastive Boundary loss** to handle severe class imbalance (<1% destroyed pixels).
- Achieved preliminary mIoU 0.37 and F1-Destroyed 0.56 on the xBD dataset (22,000+ pairs) at ~1M parameters; full A100 training in progress targeting **IEEE TGRS 2026**.

Projects

FaceGuard: Real-Time Deepfake Detection System | React, MobileNetV2, TensorFlow, FastAPI

July 2025 – Feb 2026

- Engineered a **14 MB MobileNetV2** edge-deployment model over an 88 MB XceptionNet baseline, achieving **0.04 s per-frame local CPU inference** — eliminating recurring cloud GPU costs for on-premise SME deployment.
- Achieved **83.6% test accuracy** on FaceForensics++ via transfer learning on ImageNet weights.
- Built a full-stack React web dashboard supporting static image, video streaming (SSE), and live webcam surveillance (WebSocket), deployed on Vercel/Render for zero-friction public access.
- Designed a modular preprocessing pipeline (MediaPipe ROI crop → 224×224 resize) and engineered a **temporal sliding-window buffer** to suppress false positives under motion blur.

AirChalk: Gesture-Based AI Drawing System | React, MediaPipe, Gemini Vision, HTML5 Canvas

Apr 2025 – July 2025

- Built a touchless CV system using real-time hand tracking where users draw in mid-air; Gemini Vision classifies the sketch and replaces it with an animated entity (fish, rocket, cat) — **no hardware or touch device required**.
- Implemented a **5-gesture command vocabulary** using MediaPipe landmark angle ratios (<30 ms latency), stabilised by Catmull-Rom spline smoothing to produce clean drawing trails from noisy tracking data.

Melodigram: Text-to-Music Encoder | React, Tone.js, WebGL

Jan 2025 – Mar 2025

- Developed a steganographic React application that deterministically encodes text into playable piano melodies across varying emotional registers, featuring real-time playback and key-by-key UI decoding.

Technical Skills

Edge AI & Deployment: Model optimisation (size/latency tradeoffs), CPU-only inference, Docker, Vercel, Render
Deep Learning & CV: PyTorch, TensorFlow/Keras, OpenCV, MediaPipe, Siamese Networks, U-Net, Attention Modules
Full-Stack Development: React 18, Next.js, FastAPI, REST APIs, WebSocket, SSE, Tailwind CSS, WebGL/GLSL
Languages & Data: Python (Advanced), SQL, Git, PostgreSQL, Scikit-learn, Pandas, NumPy, SciPy, EDA

Certifications & Languages

Generative AI Model Building — NxtWave (2025) | Introduction to Data Science — Simplilearn (2025) | Learning Java — Infosys Springboard (2024) | Cybersecurity Terminology — LinkedIn Learning (2024)

English: Professional proficiency

Education

B.Tech Computer Science & Engineering (Data Science)

2023 – 2027

APJ Abdul Kalam Technological University, IES College of Engineering

CGPA: 8.1 / 10